

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 4: Student Project Portfolio & Showcase App

Individual Focus: UC-04: Cast Public Vote

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1.Functional Requirements

ReqId	Type	Description	Priority	Acceptance	Rationale
FR-001	Functional	The system shall allow public participants to view projects available for public voting.	High	The system displays all projects eligible for public voting and allows a participant to select a project to view its details.	Participants must be able to review eligible projects before casting a vote.
FR-002	Functional	The system shall allow a public participant to cast a vote for an eligible project.	High	When a participant selects an eligible project and submits a vote, the system records the vote and displays a confirmation message.	Public voting is a core feature of the project showcase application.

FR-003	Functional	The system shall validate each public vote before recording it.	High	The system records a vote only if it passes the defined validation checks and rejects invalid or duplicate votes with an appropriate message.	Vote validation helps prevent duplicate or fraudulent voting and supports fair public participation.
FR-004	Functional	The system shall update the vote count of a project after a valid vote is successfully recorded.	High	After a valid vote is recorded, the selected project's vote count is updated and reflected in the project results or leaderboard.	Updating vote results ensures that public voting is accurately reflected in the showcase results.
FR-005	Functional	The system shall detect and reject invalid or suspicious voting attempts.	High	When a voting attempt fails the defined validation or security checks, the	Rejecting invalid or suspicious votes protects the integrity and fairness

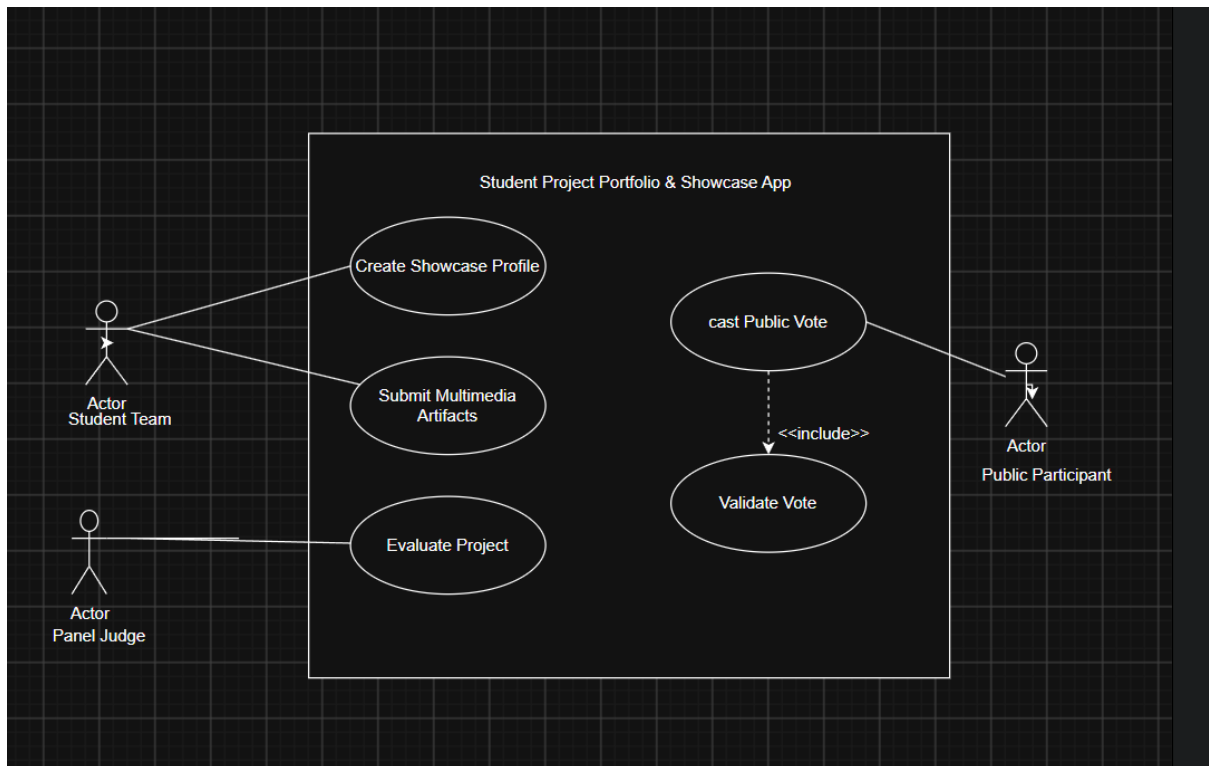
				system rejects the vote and displays an appropriate message without updating the project's vote count.	of public voting.
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2.Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance	Rationale
NFR-001	Non-Functional	The system shall ensure that public voting is protected against unauthorized and duplicate voting attempts.	High	The system prevents a participant from successfully submitting more votes than permitted and rejects unauthorized voting attempts.	Secure voting is necessary to maintain the fairness and integrity of the public voting process.
NFR-002	Non-Functional	The system shall process and confirm a valid public vote within an	Medium	Under normal operating conditions, the system shall display a vote	A responsive voting process improves user experience

		acceptable response time under normal operating conditions.		confirmation or rejection message within 3 seconds of vote submission.	and encourages effective public participation.
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3.UML Use-Case Diagram



4.Use-Case Flow Specification

Use Case ID: UC-04

Use Case Name: Cast Public Vote

Primary Actor: Public Participant

Preconditions

1. The public participant has access to the project showcase.

2. The selected project is available and eligible for public voting.
3. The public voting period is active.

Main Success Scenario:

Step

- 1 The Public Participant opens the project showcase.
- 2 The system displays the projects available for public voting.
- 3 The Public Participant selects a project and views its details.
- 4 The Public Participant selects the option to cast a vote for the project.
- 5 The system validates the voting request.
- 6 The system verifies that the vote is valid and permitted.
- 7 The system records the vote for the selected project.
- 8 The system updates the project's vote count.
- 9 The system displays a confirmation message to the Public Participant.

Postconditions

1. A valid vote is recorded for the selected project.
2. The selected project's vote count is updated.
3. The Public Participant receives confirmation that the vote was successfully submitted.

Alternate Flow A1: Duplicate or Invalid Vote Detected

A1: Invalid or Duplicate Vote Detected

- a. If the system detects that the voting request is invalid, duplicate, or does not satisfy the required validation checks, the system rejects the vote.
- b. The system does not record the vote or update the selected project's vote count.
- c. The system displays an appropriate message informing the Public Participant that the vote could not be accepted.
- d. The use case ends.